

MAJOR PROJECT REGISTRATION FORM

Date:

Project Category:	☐ Research oriented	☐ Application oriented	Batch ID:	CSM-A9
	☐ Product oriented	☐ Other (please specify)		
Specify if others:				

Title of the Project: Adaptive Benchmarking and Selection of Quantum Error Mitigation Techniques for Noisy Quantum Circuits

Abstract

Current-generation quantum computers are highly susceptible to noise and operational errors, which significantly degrade the accuracy of quantum computations. Although full quantum error correction can address these errors, its substantial qubit and hardware requirements make it challenging for near-term quantum devices. Quantum Error Mitigation (QEM) provides a practical alternative by reducing the impact of errors through statistical and post-processing techniques without requiring large-scale error-correcting codes. This project proposes an adaptive benchmarking and selection framework for evaluating quantum error mitigation techniques on noisy quantum circuits.

The study will investigate three representative QEM techniques: Readout Error Mitigation (REM), Zero-Noise Extrapolation (ZNE), and Probabilistic Error Cancellation (PEC). These techniques will be evaluated using different quantum circuit types, including Bell-state, arithmetic, and Quantum Fourier Transform (QFT) circuits, as proposed in the project specification. Experiments will initially be performed using ideal and noisy simulations with Qiskit Aer, where circuit characteristics such as number of qubits, circuit depth, gate count, two-qubit gate count, and noise levels can be systematically varied. The performance of each mitigation technique will be evaluated using metrics such as fidelity, error reduction, success probability, execution time, and computational or sampling overhead. Selected experiments will subsequently be validated using available IBM Quantum hardware, following the proposed simulator-to-hardware validation approach.

The primary research objective is to determine how circuit characteristics and noise conditions influence the effectiveness and computational cost of different mitigation techniques. The experimental results will be used to construct a benchmark dataset containing circuit, noise, mitigation, accuracy, and resource-related features. As an extension, a supervised machine-learning approach, primarily a Random Forest classifier, will be investigated for adaptive selection of REM, ZNE, or PEC based on the characteristics of a previously unseen circuit and its noise conditions. The proposed framework will therefore move beyond identifying a universally superior mitigation technique and instead investigate whether an adaptive strategy can achieve a better accuracy-cost trade-off. The expected outcome is a reproducible benchmarking methodology and adaptive decision-support framework that can provide practical guidance for selecting error mitigation techniques for near-term noisy quantum computing.

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References

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2. Koh, J. M., Koh, D. E., & Thompson, J. (2025). Readout Error Mitigation for Mid-Circuit Measurements and Feedforward. arXiv preprint arXiv:2406.07611. <https://arxiv.org/abs/2406.07611>
3. van den Berg, B., Mineev, Z. K., & Temme, K. (2023). Probabilistic Error Cancellation with Sparse Pauli-Lindblad Models on Quantum Processors. Nature Physics, 19, 1116–1121. <https://doi.org/10.1038/s41567-023-02042-2>

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